Lesson 01 Demo 02

Launching an S3 Bucket Using CloudFormation

Objective: To create an S3 bucket on AWS using CloudFormation, with a focus on defining bucket properties, securing access, and ensuring proper configuration

Tools required: AWS Management Console

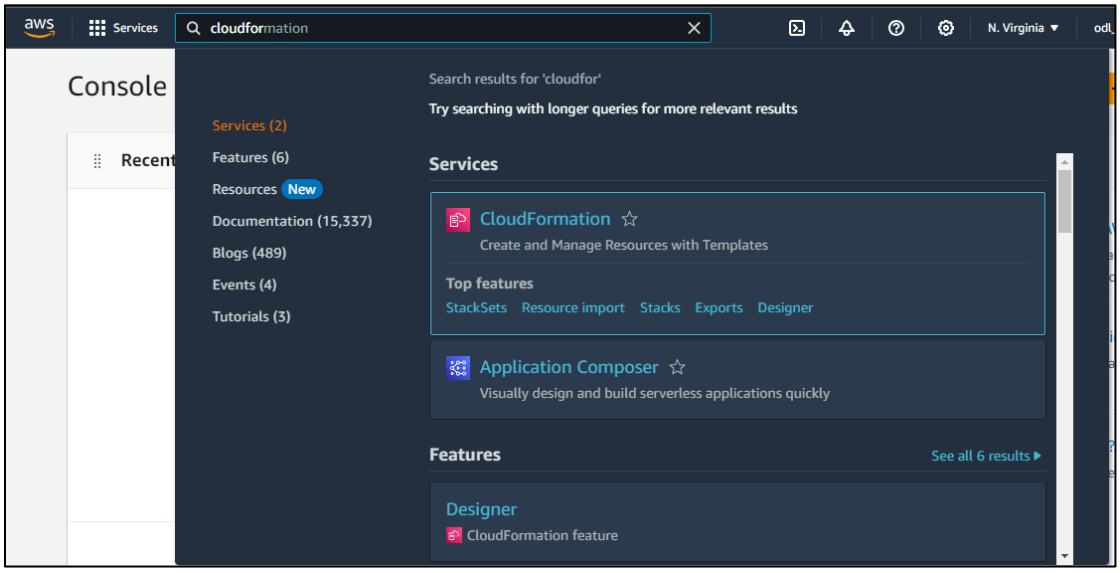
Prerequisites: None

Steps to be followed:

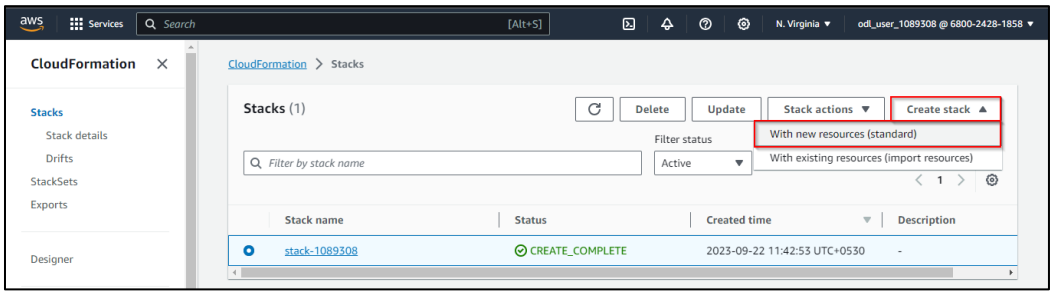
1. Create a CloudFormation template

Step 1: Create a CloudFormation template

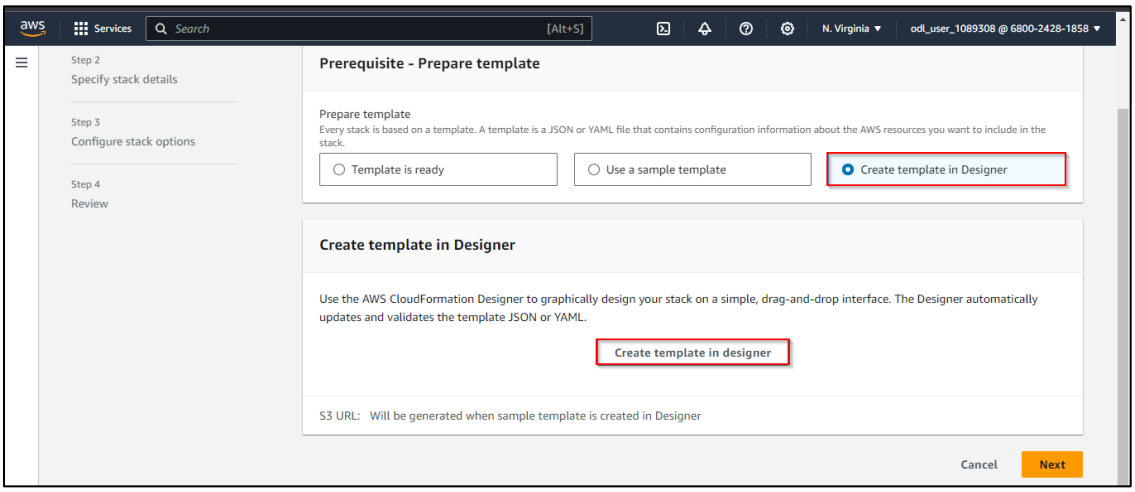
1.1 Navigate to the AWS console and search for and select **CloudFormation**



1.2 Click on **Create stack** and choose **With new resources (standard)**

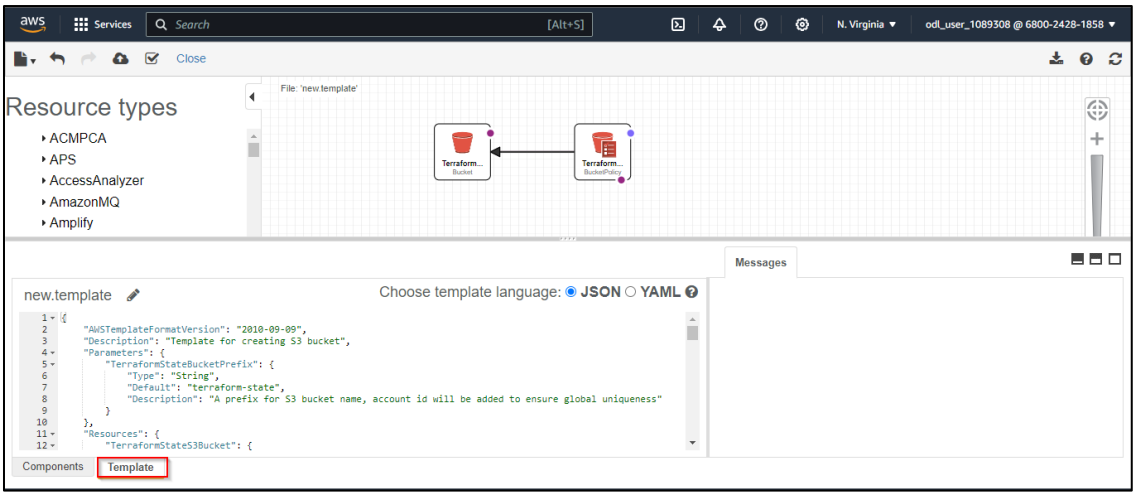


1.3 Select **Create template in Designer** to enter the template design interface

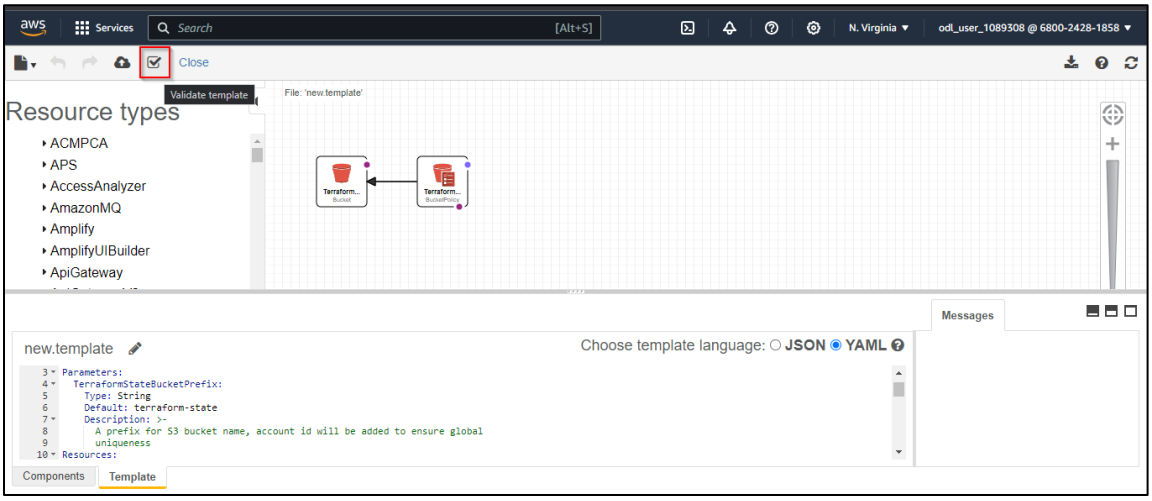


1.4 Copy and paste the following code into the **Template** section:

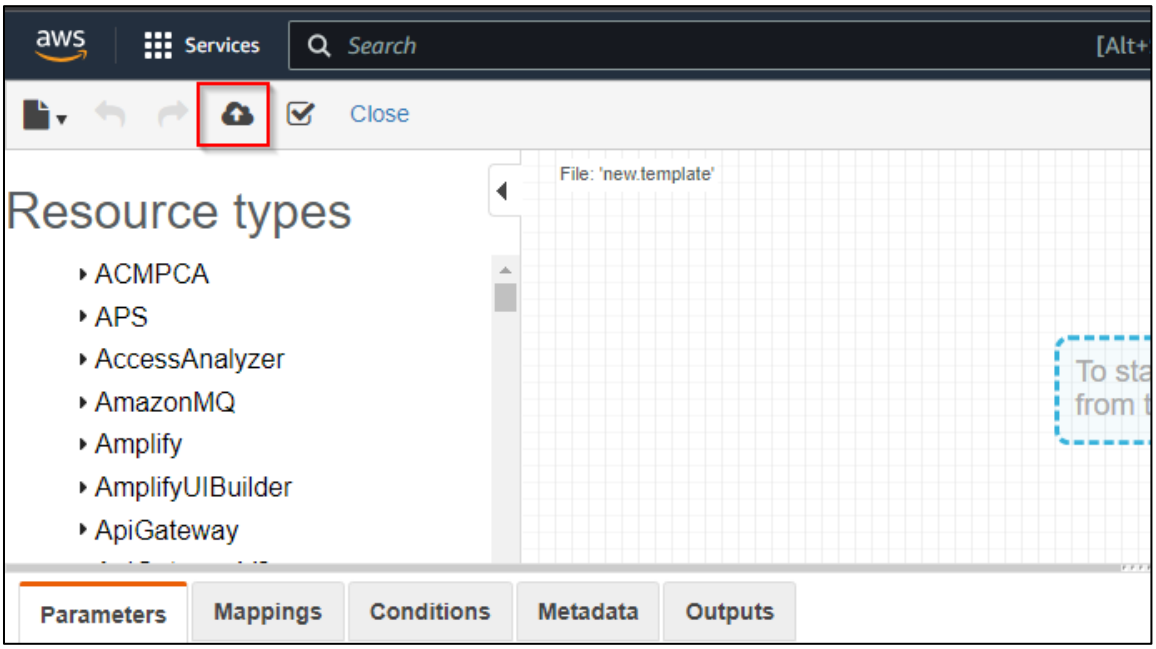
```
AWSTemplateFormatVersion: 2010-09-09
Description: 'Template for creating S3 bucket'

Parameters:
  TerraformStateBucketPrefix:
    Type: String
    Default: terraform-state
    Description: A prefix for S3 bucket name, account id will be added to ensure global
uniqueness

Resources:
  TerraformStateS3Bucket:
    Type: AWS::S3::Bucket
    Properties:
      BucketName: !Sub "${TerraformStateBucketPrefix}"
      AccessControl: Private
      PublicAccessBlockConfiguration:
        BlockPublicAcls: true
        BlockPublicPolicy: true
        IgnorePublicAcls: true
        RestrictPublicBuckets: true
      BucketEncryption:
        ServerSideEncryptionConfiguration:
          - ServerSideEncryptionByDefault:
              SSEAlgorithm: AES256
      VersioningConfiguration:
        Status: Enabled

  TerraformStateS3BucketBucketPolicy:
    DependsOn:
      - TerraformStateS3Bucket
    Type: AWS::S3::BucketPolicy
    Properties:
      Bucket: !Ref TerraformStateS3Bucket
      PolicyDocument:
        Statement:
          - Sid: DenyDeletingTerraformStateFiles
            Effect: Deny
            Principal: "*"
            Action: "s3:DeleteObject"
            Resource: !Sub "arn:aws:s3:::${TerraformStateS3Bucket}/*"
```

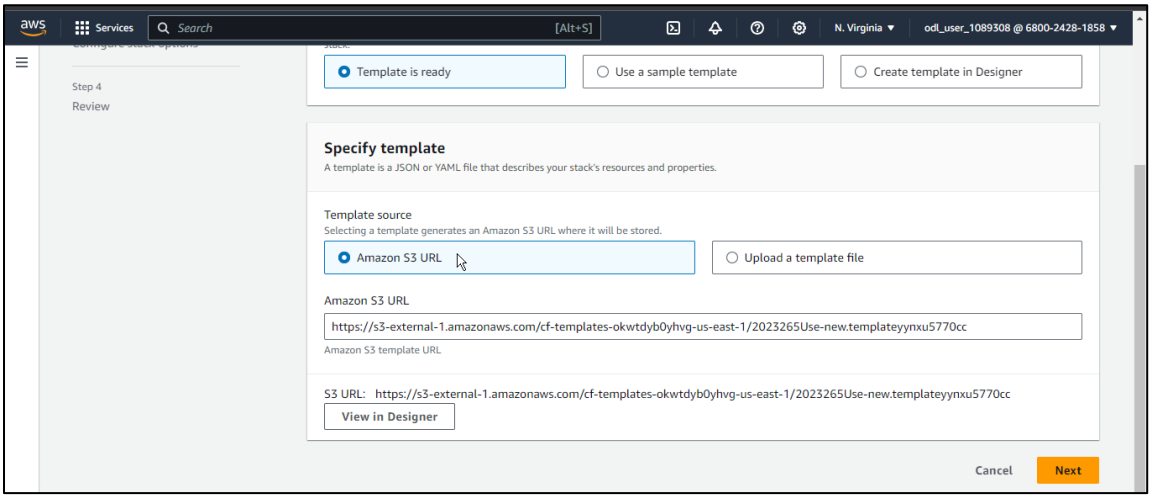


1.5 Click the **Validate template** icon to ensure the template is correct



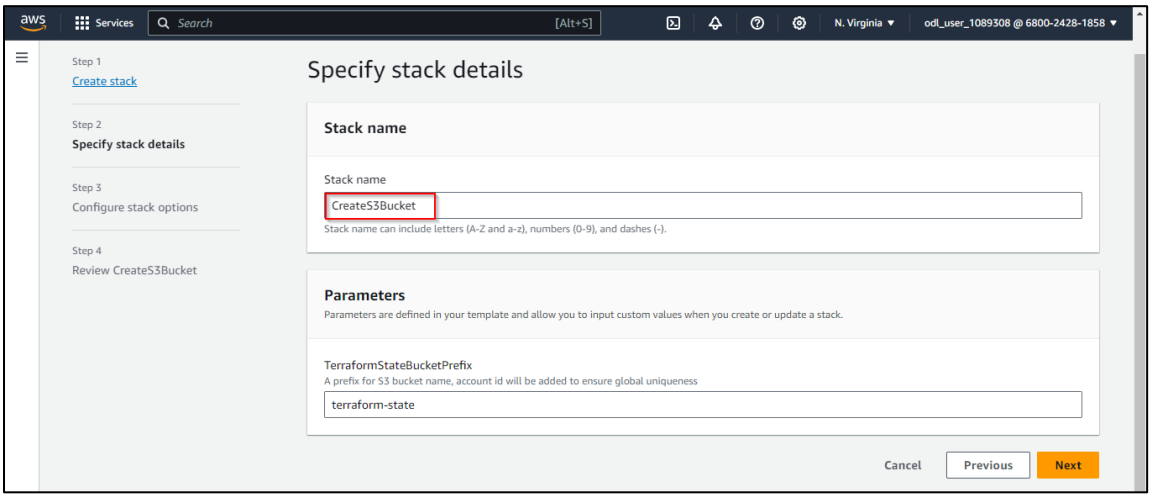
1.6 Click on the **Create stack** icon



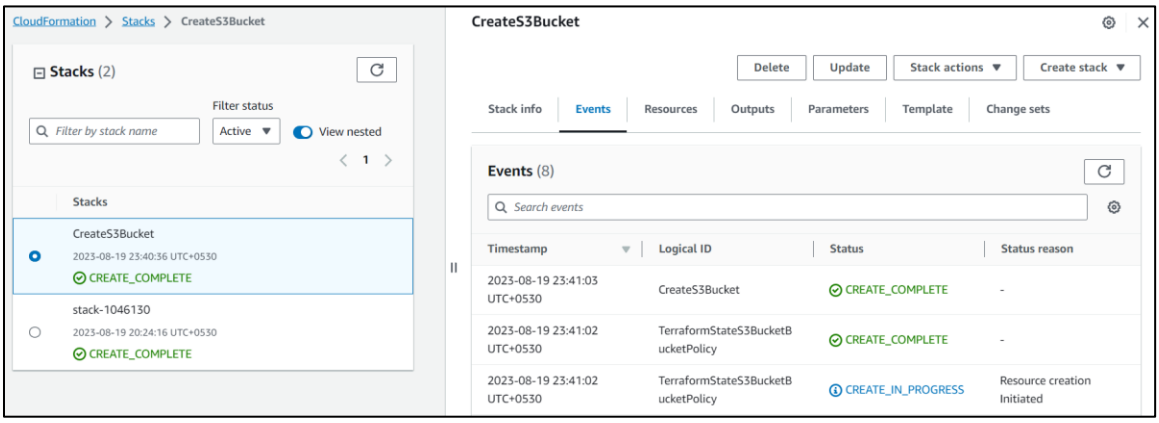


This will automatically open the **Create stack** window. Now, click on the **Next** button.

1.7 In the **Create stack** window, enter the Stack name as **CreateS3Bucket** and click **Next**

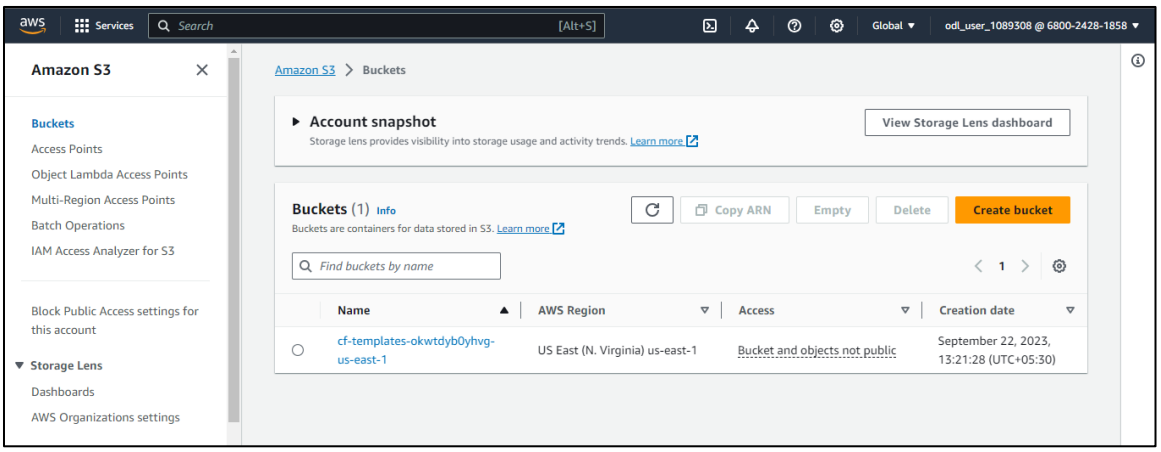


1.8 Click **Submit**



The S3Bucket has been created successfully.

After successfully completing it, navigate to the S3 service and click on **Buckets** to view the newly created bucket.



By following these steps, you have successfully created an S3 bucket with the specified configuration and access controls.